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4.1.2 Based on branching pattern and other characters

Inflorescences may also have classified based on branching, number and arrangement of flowers, and some specialized structures.

I. **Indeterminate** (**racemose**)

II. **Determinate (cymose)**

III. **Mixed inflorescence**: Inflorescence of some plants show a combination of indeterminate and determinate pattern

IV. **Special inflorescence**: Inflorescence which do not confined to these patterns

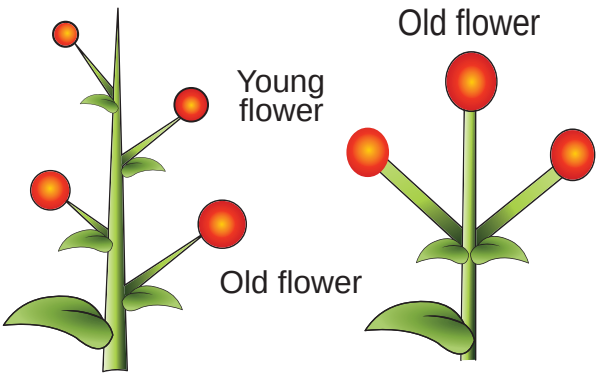


Figure 4.2: (a) Racemose

Figure 4.2: (b) Cymose inflorescence

Racemose	Cymose
Main axis of unlimited growth	Main axis of limited growth.
Flowers arranged in an acropetal succession	Flowers arranged in a basipetal succession
Opening of flowers is centripetal	Opening of flowers is centrifugal
Usually the oldest flower at the base of the inflorescence axis.	Usually the oldest flower at the top of the inflorescence axis.

I. Racemose

The central axis of the inflorescence (peduncle) possesses terminal bud which is capable of growing continuously and produce lateral flowers is called **racemose inflorescence**. Old flowers are at the base and younger flowers and buds are towards the apex. It is further divided into 3 types based on growth pattern of main axis.

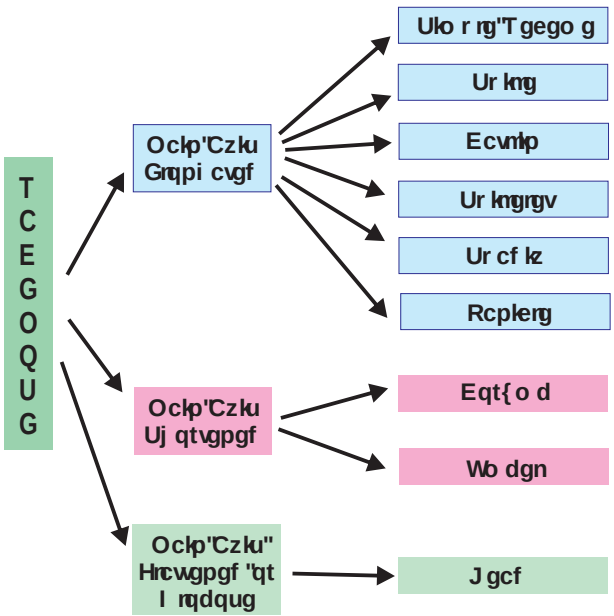


Figure 4.3: Racemose

1. Main axis elongated

The axis of inflorescence is elongated and contains pedicellate or sessile flowers on it. The following types are discussed under main axis elongated type.

a. Simple raceme: The inflorescence with an unbranched main axis bears **pedicellate flowers** in acropetal succession. Example: *Crotalaria retusa,* mustard and radish.

b. Spike: Spike is an unbranched indeterminate inflorescence with **sessile flowers**. Example: *Achyranthes, Stachytarpheta.*

c. Spikelet: Literally it is a small spike. The Inflorescence is with